Spontaneous DNA Formation and the First Life on Earth

Jeremy Favro (0805980)
Trent University, Peterborough, ON, Canada

November 5, 2026

What am I talking about

- DNA is made by copying other DNA. RNA is half of DNA
- Bootstrap paradox: where did the first DNA come from?
- How can we replicate these conditions?

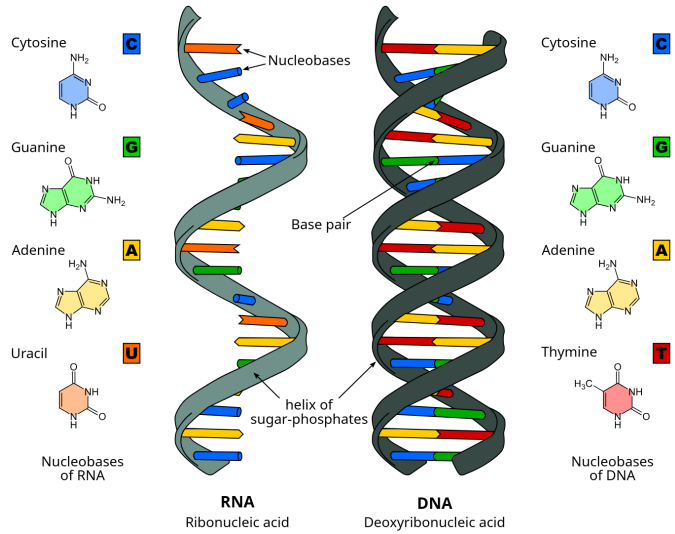


Figure: RNA/DNA structures.
wikipedia.org/wiki/Nucleic_acid

Thermodynamic/Energetic Barriers

- Spontaneous organization of *nucleotides* is limited energetically and physically
- Want to bond phosphate-to-phosphate, but they may not be pointed at each other
- Also *endergonic*, $\Delta G > 0$
- Also anti-entropic ($\Delta S < 0$), combining molecules increases order

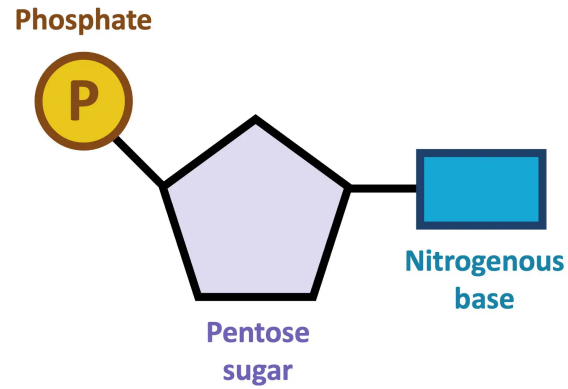


Figure: A nucleotide.

Encouragement

- Maybe heat? Temperatures required would destroy the nucleotides
- Reagents? Unlikely to be responsible for proto-RNA; highly complex chemically
- To thermodynamics!

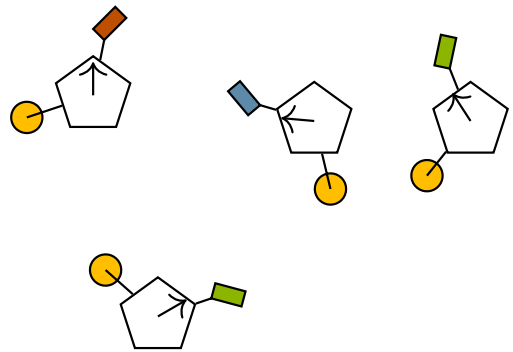


Figure: A bunch of nucleotides whizzing around.

Constraint

- We pre-organize the molecules
- Force them onto the surface of a sphere
- Increases the Gibbs free energy

$$G = U + PV - TS$$

by decreasing degrees of freedom

- Our problem of $\Delta G > 0$ is lessened



To create a rabbit out of nothing and place it on the table, the magician need not summon up the entire enthalpy, $H = U + PV$. Some energy, equal to TS, can flow in spontaneously as heat; the magician must provide only the difference, $G = H - TS$, as work.

Figure: Cartoon regarding the Gibbs free energy by Daniel Schroeder

Constraint

- Levitating spheres of nucleotide containing water
- Creates a spontaneous reaction, allowing nucleotides to bond
- Entirely possible in early-Earth conditions; raindrops, wave spray

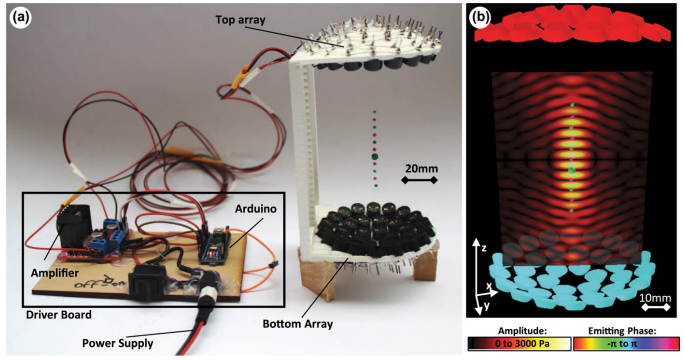


Figure: An ultrasonic levitator.

Summary

- Nucleotide bonding into DNA/RNA energetically unfavourable, physically unlikely
- Constrain nucleotides to a 2D surface
- Enables spontaneous bonding and DNA/RNA formation
- Ultrasonic levitators are much less complex than other solutions

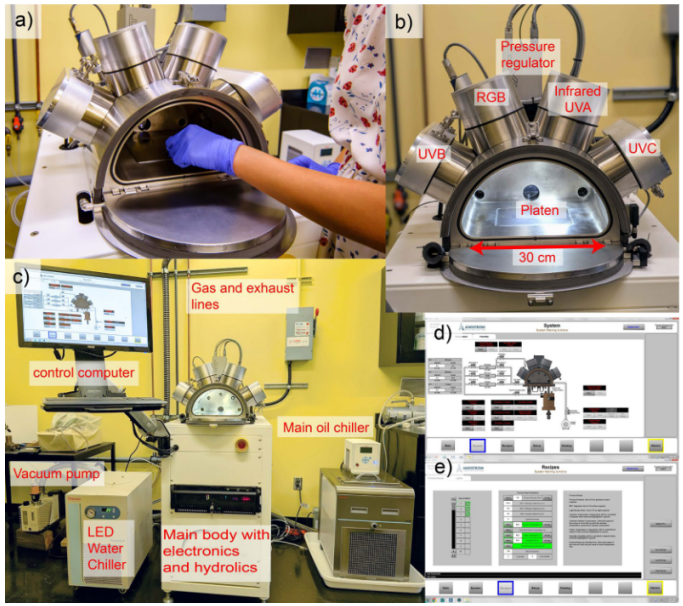


Figure: Non-levitating wet-dry cycle device.

[1] Povilas Simonis et al. "Abiotic synthesis of RNase-resistant phosphodiester and pyrophosphate-linked polymers via thermodynamically controlled wet-dry cycling". In: *Scientific Reports* 16.1 (June 2026), p. 26937. ISSN: 2045-2322. DOI: 10.1038/s41598-026-56548-9. URL: https://doi.org/10.1038/s41598-026-56548-9.